System Validation Report

Online Bookstore – SE Mini-Project (UE24CS341A), PES University, Group 2

Part 3 – Testing (2–6 November 2026)

Item	Detail
Project	Online Bookstore (multi-seller book marketplace)
Prepared by	Group 2
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Date	6 November 2026
Repository	SE-OnlineBookstore/SE-Mini-Project
Version	1.0

1. Purpose and scope

This report records the system validation of the Online Bookstore against the Test Plan (TC-01 to TC-16). Each case was executed against the running application (React client and Express/MongoDB server, seeded with 2,500 titles and 10,000 listings) and evidenced with screenshots kept in the evidence folder.

2. Environment

Item	Detail
Server	Node.js, Express 4, Mongoose 8, MongoDB
Client	React 18 + Vite 5
Data	Seeded accounts buyer1–4, seller1–4, admin1–4; 2,500 titles; 10,000 listings
Machine	Single Windows laptop, local servers
Mail	Stub (OTP written to server log)
Load tool	autocannon (TC-13)

3. Summary of results

Measure	Count
Test cases in plan	16
Executed	16
Passed	16
Failed (final)	0
Pending evidence (assigned to Test Engineer)	0
Defects raised / closed	2 / 2 (DEF-01, DEF-02)

4. Execution results

TC ID	Title	Date	Tester	Result
TC-01	Registration with valid data	2026-11-03	Deeptha	Pass
TC-02	OTP verification	2026-11-03	Deeptha	Pass
TC-03	Login and token handling	2026-11-03	Deeptha	Pass
TC-04	RBAC unauthorized access	2026-11-03	Deeptha	Pass
TC-05	Catalogue pagination	2026-11-03	Deeptha	Pass
TC-06	Search and filtering	2026-11-03	Deeptha	Pass
TC-07	Verified-purchase review	2026-11-05	Deeptha	Pass
TC-08	Cart quantity and persistence	2026-11-05	Koushik	Pass
TC-09	Checkout and payment success	2026-11-05	Koushik	Pass
TC-10	Order cancellation/return	2026-11-05	Deeptha	Pass
TC-11	Seller listing/inventory	2026-11-05	Deeptha	Pass
TC-12	Administrator moderation/reporting	2026-11-05	Deeptha	Pass
TC-13	Search performance	2026-11-05	Deeptha	Pass
TC-14	Security validation suite	2026-11-05	Deeptha	Pass
TC-15	Responsive/accessibility/browser compatibility	2026-11-05 (retest 2026-11-06)	Koushik	Pass
TC-16	Concurrent last-stock purchase	2026-11-05	Deeptha	Pass

5. Detailed notes and evidence

TC ID	Evidence files	Observations
TC-01	TC-01_a.png TC-01_b.png	Account created; six-digit OTP sent (mail stub log)
TC-02	TC-02_a.png TC-02_b.png	Wrong code rejected; correct code verified; signed in as Buyer2
TC-03	TC-03_a.png TC-03_b.png TC-03_c.png	Access token 15 min; refresh cookie set; refresh 200; logout 204; old refresh token 401
TC-04	TC-04_a.png TC-04_b.png	Buyer got 403 FORBIDDEN_ROLE on seller and admin routes; ACCESS_DENIED audit entries written with matching request IDs
TC-05	TC-05_a.png TC-05_b.png	20 items per page; page 1 of 125 (2500 books); summary fields present; page 2 returns 20
TC-06	TC-06_a.png to TC-06_f.png	Title/author/ISBN/publisher search and filters correct; single search 0.12s (load test under TC-13); observation: search box keeps old text after the

		search chip is removed (cosmetic)
TC-07	TC-07_a.png TC-07_b.png TC-07_c.png	Review rejected without a delivered purchase of that book record; accepted after delivery (rating 4, count updated); second review for the same order line refused.
TC-08	TC-08_a.png to TC-08_k.png (TC-08-b.png)	Quantity limited by available stock and 1 to 10; cart persisted after logout and login.
TC-09	TC-09_A.png to TC-09_F.png	Order confirmed only after successful payment; stock reduced accordingly.
TC-10	TC-10_a.png to TC-10_e.png	Cancel before dispatch accepted; cancel unavailable after dispatch; return accepted within 7 days of delivery.
TC-11	TC-11_a.png to TC-11_k.png	Initial run 4 Nov: DEF-01 MRP not enforced on create (Rs 99999 accepted). Fixed in PR 87. Retest 5 Nov: create above MRP rejected. Edit, MRP on edit, low-stock alert, out-of-stock and back, accept-pack-dispatch-deliver all pass.
TC-12	TC-12_a.png to TC-12_d.png	Admin blocked a listing with a reason : listing hidden from buyer view, status BLOCKED for seller; block and unblock recorded in audit log with reasons; monthly report exported as CSV and PDF for 2026-11.
TC-13	TC-13_a.png TC-13_b.png	200 concurrent connections for 20 s on 10,000 seeded listings, 32k requests, 0 failures: p97.5 234 ms, p99 282 ms (limit 2000 ms). Text index used in the query plan. Per-IP rate limit (SR-10) off via NODE_ENV=test and request logging off for the run; one repeated query so results largely cached; single-machine run.
TC-14	TC-14_a.png to TC-14_d.png	Login lockout after 5 failures; global limit 100 requests/min (429 after); no/tampered token 401;

		invalid input, ADMIN self-registration, extra fields and NoSQL operator all 400; webhook without/with wrong signature 401; ACCESS_DENIED, STOCK_EXHAUSTED and payment events present in audit log. HSTS and security headers present; TLS itself not testable on localhost.
TC-15	TC-15_a.png (before fix), TC-15_b.png (after fix)	Initial run 5 Nov: Add to cart required sideways scrolling at 360 px (DEF-02). Fixed in PR 91. Retest 6 Nov: button visible, Pass.
TC-16	TC-16_a.png TC-16_b.png	Two buyers placed orders for the last copy and paid concurrently: exactly one order confirmed, the other got INSUFFICIENT_STOCK with refund pending; stock ended at 0 (not negative). First attempt interrupted by the dev auto-reloader; rerun with npm start.

6. Defects

ID	Found in	Description	Status
DEF-01	TC-11 (4 Nov)	Listing creation did not enforce MRP (a price of Rs 99999 was accepted).	Fixed in PR #87; retested 5 Nov – Pass; new automated test added
DEF-02	TC-15 (5 Nov)	Add to cart and wishlist buttons in the book offers table were off-screen on small screens and needed sideways scrolling at 360 px.	Fixed in PR #91; retested 6 Nov – Pass

7. Limitations

- TC-12 was verified through the API because the admin page (S3-04) was not merged at test time.
- TC-13 ran with the per-IP rate limit (SR-10) off and request logging off; the load used one repeated query, so results are largely cached; single-machine run. Result: 200

connections for 20 s, about 32,000 requests, 0 failures, p97.5 234 ms, p99 282 ms against a 2,000 ms limit.

- TC-14: TLS cannot be tested on localhost; HSTS and security headers were confirmed present.
- Observation (cosmetic, not a defect against requirements): the search box keeps the old text after the search chip is removed.

8. Conclusion

All 16 test cases passed. Two defects were found, fixed and retested (DEF-01 and DEF-02).